TECHNICAL DESCRIPTION

Skills Challenge 2026

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Skill: Autonomous Mobile Robotics

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INTRODUCTION

THE NAME OF THE SKILL COMPETITION IS

Autonomous Mobile Robotics

DESCRIPTION OF THE ASSOCIATED WORK ROLE(S) OR OCCUPATIONS(S)

Mobile Robotics is a rapidly evolving industry with applications in diverse sectors such as autonomous vehicles, manufacturing, agriculture, aerospace, mining, and medicine. Robotics technologists, playing a crucial role, work in offices, manufacturing plants, or laboratories, designing, maintaining, and developing new applications for robots. They address specific business problems, like automating tasks in manufacturing and designing robots for inaccessible or hazardous areas. The process involves client consultation, specification development, design, prototype assembly, programming, and testing for consistent performance. Robotics technologists need expertise in logic, microprocessors, programming, and mechanical, electrical, and control systems.

Key responsibilities include preparing specifications, ensuring cost-efficient designs, cost-price calculations, and quality control. High-performing robotics technologists possess organizational and self-management skills, effective communication, interpersonal abilities,

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teamwork, and innovation in problem-solving. They must adapt to working internationally across various sectors, transferring analytical skills, committing to ongoing professional development, and embracing experimentation and risk-taking within boundaries. The growing global robotics industry offers sustainable career opportunities, requiring collaboration with diverse cultures and adapting to rapid technological changes. The skill set associated with robotics technologists is expected to broaden.

NUMBER OF COMPETITORS PER TEAM

Autonomous Mobile Robotics is a team skill competition with two Competitors per team.

THE TEST PROJECT

The Test Project is a series of 3 modules.

MODULE	NAME	APPROX. HOURS	PLACE
1	Work Organization and Behaviour	3 h	Competition Area
2	Robot Design	30 min	Workstation
3	Performance Review and Commissioning	45 min	Court
TOTAL		3 h	

Note: The work organization and behavior module will be evaluated during the 3 hours of competition. The design and robot performance module will be evaluated at a specific moment according to the timetable and during the same 3 hours of competition.

MODULE 1 – WORK ORGANIZATION AND BEHAVIOR

The individual needs to know and understand:

- Principles and applications of safe working generally and about manufacturing
- The purposes, use, care, and maintenance of all equipment and materials, together with their safety implications
- Environmental and safety principles and their application to good housekeeping in the work environment
- Principles of teamwork and their applications
- Personal skills, strengths, and needs relative to the roles, responsibilities, and duties of others individually and collectively
- The parameters within which activities need to be scheduled

The individual shall be able to:

- Prepare and maintain a safe, tidy, and efficient work area
- Prepare himself for the tasks on hand, including full regard to health and safety

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- Schedule work to maximize efficiency and minimize disruption
- Take account of the rules and regulations in force for robotics technician/engineering
- Select and use all equipment and materials safely and in compliance with manufacturers' instructions
- Apply or exceed the health and safety standards applied to the environment, equipment, and materials
- Restore the work area to an appropriate state and condition
- Contribute to team performance both broadly and specifically
- Give and take feedback and support

MODULE 2 – ROBOT DESIGN

The individual needs to know and understand:

- The principles and applications of project design
- The nature and formats of project specifications
- The basis on which the manufactured item will be appraised
- Design parameters include the following:
  - Options appraisal
  - Selection of components, materials, and work processes
  - Prototype development
  - Manufacture
  - Assembly
  - Refinement
  - Commissioning
- Principles and applications for:
  - Designing, assembling, and commissioning mobile robotics systems
  - The components and functions of electrical and electronic systems
  - The components and applications of add-ons
  - The components and applications of mobile robotics systems
- Principles and applications of design and assembly of mechanical, electrical, and electronic systems, their standards, and their documentation
- Principles and methods for work organization, control, and management concerning the product
- Principles and techniques for generating creative and innovative solutions

The individual shall be able to:

- Analyze the briefs or specifications to identify the required performance characteristics of mobile robots
- Identify and resolve areas of uncertainty within the briefs or specifications
- Identify the characteristics of the environment in which the mobile robots are required to operate
- Identify hardware requirements to support the mobile robots' performance
- Generate designs for the manufacture of a functioning item within given timescales

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- Generate designs for teleoperation control systems independent of the base unit
- Develop strategies to solve mobile robotics tasks including navigation and orientation
- Generate innovative solutions to design challenges
- Identify and appraise options for the selection, purchase, and manufacture of materials, components, and equipment

MODULE 3 – PERFORMANCE REVIEW AND COMMISSIONING

The individual needs to know and understand:

- Criteria and methods for testing equipment and systems
- Criteria and methods for operating test runs
- The scope and limits of the technologies and methods used
- Strategies for thinking creatively and generating innovation
- The possibilities and options for making incremental and/or radical changes

The individual shall be able to:

- Test each part of a mobile robot against the agreed operating criteria
- Test mobile robots' overall performance against agreed operating criteria
- Optimize the operation of each part of the systems, and the systems as a whole, through analysis, problem-solving, and refinement
- Undertake final test runs to commission systems
- Review each part of the process of design, fabrication and assembly, and operation, against established criteria, including accuracy, consistency, time, and cost
- Ensure that all aspects of a design stage meet the required industry standards
- Finalize and present portfolios to clients, the portfolios to include all essential documentation required in business transactions
- Present mobile robots and portfolios to clients and respond to questions

TEST PROJECT DEVELOPMENT AND CIRCULATION

The Test Project/modules are developed by the Chief Expert.

TIME	ACTIVITY
Three (4) months before the Competition	The Pre Test Project is developed and presented to Experts, Teachers, and Competitors
After the release of the Pre Test Project	Training of Experts, Teachers, Competitors for the test project
One (1) day before the Competition	The Main Test Project is changed if applicable
At the Competition	Executing the test project

The Test Project is circulated via MS Teams by the Chief Expert.

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THE MARKING SCHEME

The Marking Scheme and Test Project will be developed by the Chief Expert, who is free to define the criteria that they consider to be the most suited to the assessment and marking of the Test Project.

ASSESSMENT AND MARKING USING JUDGMENT

Judgement-based criteria assessment will not be conducted during the competition due to the absence of a panel of three or four Experts to support the evaluation process. Therefore, all criteria included in the marking scheme will be defined and applied exclusively as measurement criteria.

ASSESSMENT AND MARKING USING MEASUREMENT

Assessment will be conducted by the Chief Expert responsible for evaluating all aspects. The Expert will ensure consistency and accuracy throughout the assessment process. Unless otherwise stated, only the maximum mark or zero will be awarded. Where applicable, criteria for partial marks will be clearly defined within each aspect.

SKILL ASSESSMENT STRATEGY

SECTION	CRITERION	JUDGMENT MARKS	MEASUREMENT MARKS	TOTAL
A	Work organization and management	0.00	10.00	10.00
B	Robot Design	0.00	10.00	10.00
C	Performance and Commissioning	0.00	80.00	80.00
TOTAL		0.00	100.00	100.00

CRITERION A – WORK ORGANIZATION AND MANAGEMENT

- Team Space Condition during all Competitions Day.
- Adherence to schedules during all Competitions Day.
- Co-operative Behavior with Compatriot Team Members, Opponent Competitors, and Experts during all Competitions Day.

CRITERION B – ROBOT DESIGN

- Evaluation of the Robot's design is divided into two main sections:
    - Base Robot:

- Mechanical
- Electrical
- Sensors
- Operation

- Object Management System:
  - Mechanical
  - Electrical
  - Sensors
  - Operation

CRITERION C – PERFORMANCE AND COMMISSIONING

- Test Project Run is the evaluation of the performance of the teams' robots on the Main Test Project Task.
- There will be 2 separate runs, each one evaluated independently.
- The layout of the environment will be given during the briefing period.
- Competitors will be given time to program the task.
- The performance of the Robot may be evaluated by the ability of the robot to: Communicate states using visual indications (lights, LEDs), retrieve target objects, transport target objects, deliver target objects, exit specified locations, enter specified locations, not crash into walls/objects, and time taken (only if 100% of the task is completed);

SKILL ASSESSMENT PROCEDURES

The Chief Expert is responsible for conducting the assessment process, ensuring that all evaluation activities are carried out consistently and in accordance with the defined marking scheme.

SECTION	CRITERION	TEST END OF DAY	MIN. NUMBER OF TEST GROUPS
A	Work organization and management	C1	Chief Expert
B	Design	C1	Chief Expert
C	Performance and Commissioning	C1	Chief Expert

SAFETY REQUIREMENTS

- The workstation must be kept in safe and clean working conditions.
- Any irregularity during non-compliance with the safety rules, the appropriate penalties will be taken.

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- If a competitor is not using the appropriate safety equipment, the Expert will have to inform and guide it to be used.
- Individual safety equipment is the responsibility of the Competitor.

TASK	SAFETY GLASSES WITH SIDE PROTECTION	CUT PROTECTION GLOVES	STURDY SHOES WITH CLOSED TOE AND NO HEEL	HEARING PROTECTION	DUST MASK
General PPE for safe areas			X		
Material manipulation such as drilling, cutting and grinding	X	X (optional for material removal only)	X	X	X
Soldering	X	X (optional for material removal only)	X	X	X
General work at workstation			X		

MATERIAL AND EQUIPMENT

The Infrastructure List details all equipment, tools, materials, and facilities that will be provided by the school for both the training of competitors and the day of the competition. The Competition Organizer will not provide materials and equipment for training.

Competitors can use their own toolbox with a total external volume not exceeding 0.06 m^3 (Volume = Length x Height x Width – Example: 60 cm × 40 cm × 25 cm). Volume measurement does not include a packing crate, other protective packing material, palette for transportation, or wheels.

Competitors are not allowed to use power tools. Electric screwdrivers are the only exception.

Competitors may manufacture and create custom components for their mobile robots. Some restrictions will be enforced and checked before the Competitors may build their robots. Further information and details will be provided in the Test Project circulated (TP).

Furthermore, Competitors are required to supply their own Personal Protective Equipment as specified in the previous section safety requirements.

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SKILL SPECIFIC RULES

TOPIC/TASK	SKILL-SPECIFIC RULES
Use of technology – USB, memory devices	• Competitors are allowed to bring one USB memory stick per • Competitor team into the workshop on the Familiarization Day. During competition days, the USB memory devices cannot leave the competition floor and must be kept in a closed envelope with the Workshop Manager to be used only in exceptional cases to recover a critical state. It can be removed from the workshop at the end of the competition on C4 only. • Experts and Interpreters are not allowed to bring USB memory devices into the workshop. • The Skill Competition Manager and Chief Expert can bring USB memory devices for competition organization purposes only. They can be removed from the workshop at the end of the competition on C4 only. • Additionally, Competitors are allowed to bring one Micro SD Card to be used on the VMX per Competitor team into the workshop on Familiarization Day. During the competition, the Micro SD Card needs to remain inside the workshop, together with the VMX. No extra Micro SD Card can be used by Experts, Interpreters, SCM, or CE. In the case of critical failure, the USB memory stick can be used to recover the Micro SD or a generic Micro SD can be provided by the Sponsor (as spare parts).
Use of technology – personal laptops, tablets, and mobile phones	• Competitors are allowed to bring one laptop into the workstation. • Competitors are not allowed to bring tablets into the workstation. • Experts are not allowed to bring laptops or tablets into the workshop. If these items are brought, they must be locked in the personal locker and can only be removed at the end of the day. The Skill Competition Manager and Chief Expert are exempt from this rule. • Competitors are not allowed to bring mobile phones into the workshop. If these items are brought, they must be locked in the personal locker and can only be removed at lunchtime and at the end of the day. • Skill Competition Manager, Chief Expert, Experts, and Interpreters are allowed to bring mobile phones into the workshop and they can remain in their pocket but shouldn't be used in public except for emergencies.
Use of technology – personal photo and video taking devices	• Skill Competition Manager, Chief Expert, Competitors, Experts, and Interpreters are allowed to use a personal photo, video-taking devices, and wearable devices in the workshop outside competition times only. This rule applies C-4 until C+1.
Templates, aids, etc.	• Competitors are allowed to bring and use aids for positioning their robot. • Competitors are allowed to bring and use manipulator tools to hold the robot during assembly and wiring.

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Drawings, recording information	• Teams can bring all kinds of information on paper and/or digitally. This information may be brought into the workshop on Familiarization Day only and kept in the workshop for the duration of the competition. • Competitors may not gather new information during the competition over the Internet and/or by consultation.

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